# Replace with Difference

**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given an array of $n$ integers. You will perform $n - 1$ operations on the array.

In one operation, you will choose two numbers $a$ and $b$ from the array, delete both of them from the array and add $|a - b|$ into the array.

Your task is to find a sequence of operations such that the last number remaining in the array is $0$.

## Input

The first line has an integer $n$: the length of the array.

The next line has $n$ integers $x_1, x_2, \dots, x_n$: the contents of the array.

## Output

Print $n - 1$ lines each containing two integers $a$ and $b$: the numbers chosen in the operations. You can print any valid solution.

If no solution exists, print only $-1$.

## Constraints

- $2 \le n \le 1000$
- $1 \le x_i \le 1000$

## Example

| Input | Output |
|---|---|
| 5<br>2 7 4 12 1 | 2 12<br>7 10<br>4 1<br>3 3 |

*Explanation*: The array changes as follows:

- $[2, 7, 4, 12, 1] \rightarrow$ remove $2$ and $12$, add $10$
- $[7, 4, 1, 10] \rightarrow$ remove $7$ and $10$, add $3$
- $[4, 1, 3] \rightarrow$ remove $4$ and $1$, add $3$
- $[3, 3] \rightarrow$ remove $3$ and $3$, add $0$
- $[0]$: the final array